

Shared evidence | Constructed systems models

Animal systems: the evidence challenge

Name _____ Date _____

All records below are invented teaching data. They do not describe pupils or diagnose health.

Pump	Heart rate (beats/min)	Stroke volume (mL/beat)
A	60	70
B	90	60

Pump output = heart rate × stroke volume. There are 1,000 mL in 1 L. Neither oxygen content nor tissue demand was measured.

Exchange case	Relative area	Relative concentration difference	Relative thickness
X	3	4	1
Y	6	4	2
Z	6	4	1

Relative diffusion index = area × concentration difference ÷ thickness, using the same omitted constant. The rule is idealised and has no calibrated clinical unit.

Control case: glucose absorbed from the gut has raised blood glucose. Pancreatic beta cells release more insulin. Responsive tissues increase uptake and liver/muscle storage is promoted. This is a qualitative model, not treatment advice.

W1. Choose the area-only pair. W2. Choose the thickness-only pair. W3. Repair:
“B has a higher output, so every tissue definitely gets enough oxygen.”

Supported route | Build an evidence note

Use the shared page. Sentence starters: “The calculation shows...”; “This comparison keeps...”; “It does not show...”.

S1. Calculate outputs for A and B in mL/min and L/min. Which is greater?

S2. Choose the area-only pair from X, Y and Z. Name the two unchanged factors and predict the relative effect.

S3. Explain how insulin differs from glucose and glycogen in the control case.

S4. Repair: “A higher pump output proves enough oxygen reaches every cell.” Keep what is supported and state one missing measurement.

Standard route | Audit three claims

Use mechanisms and data in every answer where relevant.

M1. Calculate A and B outputs in L/min and the difference. Explain why comparing heart rate alone is unsafe as a general shortcut.

M2. Calculate indices for X, Y and Z. Explain why X and Y do not isolate an area effect.

M3. Explain the insulin response to raised blood glucose, distinguishing the hormone signal from the stored material.

M4. Trace oxygen from alveolar air to a tissue cell. Use this pathway to explain two limits of inferring oxygen adequacy from pump output alone.

Stretch route | Design the next comparison

Optional extension uses the same model; no Higher-only endocrine mechanism is required.

T1. A pump C has 120 beats/min and 35 mL/beat. Compare its output with A and B. What does this show about using heart rate alone?

T2. Design an area-only comparison with X, using an area of 9. State the concentration difference, thickness and predicted relative index.

T3. The supplied model omits oxygen content and tissue demand. Explain how either could change the adequacy judgement even if output stayed the same.

T4. Write a concise advice note to the model's designer: one supported conclusion, one unsupported claim and one specific extra measurement.

Exit and reflection | Evidence before certainty

Answer independently or explain to an adult who records your reasoning.

E1. Calculate $80 \text{ beats/min} \times 50 \text{ mL/beat}$ in L/min.

E2. Why is X/Y not an area-only test?

E3. Name one missing quantity needed to judge whether oxygen supply meets tissue demand.

L1. Record the audience's evidence question and the revision it prompted.

My next step

- A. Keep calculation units consistent.
- B. Choose a controlled comparison.
- C. Separate signal from material.
- D. State what evidence cannot establish.